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DEGAS-Analysis of organic matter degradation during weathering of Paleozoic black shales

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Fig. 1: Recent weathering on a outcrop of Silurian alumn shales

INTRODUCTION

Overmature Silurian black shales with an average $C_{org.}$-content of 0.01 to 20 wt. % were investigated with regard to their different degree of weathering (Fig. 1 & 2). Specimens affected by Mesozoic - Tertiary weathering showed deep bleached zones and black carbon-rich inner cores. Due to the high thermal maturity usual organic geochemical methods based on solvent extracts are not applicable to these shales. Directly coupled TG-MS allows to measure certain pyrolysis parameters for investigation of weathering processes. Small amounts of sampling material (10 mg) allow studies in mm-scale.

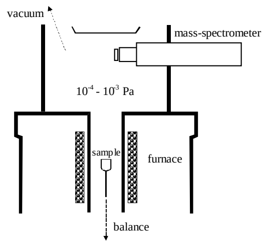
Fig. 2: Mesozoic - Tertiary weathering caused deep bleaching

Fig. 3: Location of sampling

Fig. 4: 1 - Alumn shale / lydite layers – recent weathering
2 - Same formation – Mesozoic-Tertiary weathering
3 - Bleached alumn shales with dark $C_{org.}$ rich cores

METHOD AND SAMPLES

Combined TG/MS techniques have been used widely for the simultaneous detection of organic and inorganic volatiles in pyrolysis experiments. One major problem of these methods is the possibility of reverse reactions of evolved volatiles. To avoid this we used the pyrolysis technique DEGAS, consisting of a TG coupled directly to a quadrupole mass spectrometer (Heide et al. 2000, Fig. 5). Vacuum pyrolysis combined with direct mass spectrometric detection (no pressure reduction) has the advantage of highly nonequilibrated conditions without the possibility of reverse reactions of the evolved volatiles. The system enables a temperature resolved analysis of hydrocarbons (comparable to RockEval) and the simultaneous detection of inorganic volatiles (e. g. H_2O, CO_2, CO, N_2, SO_2) during heating in vacuum (Schmidt & Heide 2001).
For this study we used whole rock samples of Silurian black shales affected by recent (Fig. 4.1) and Mesozoic - Tertiary weathering (Fig. 4.2).



Directly
coupled
Evolved
Gas
Analyzing
System

Fig. 5: DEGAS, directly coupled evolved gas analyzing system

RESULTS

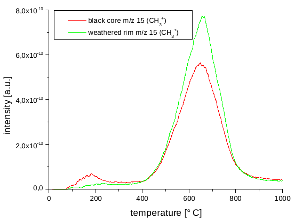


Fig. 6: Pyrolysis of organic matter illustrated by m/z 15

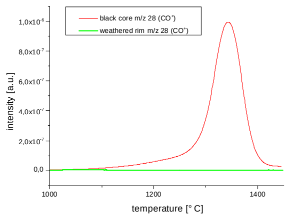


Fig. 7: Oxidation of the remaining organic matter and formation of CO at high temperatures

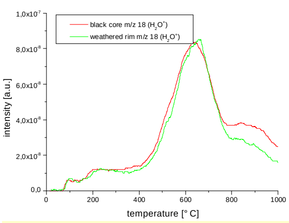


Fig. 8: Water release during pyrolysis (m/z 18)

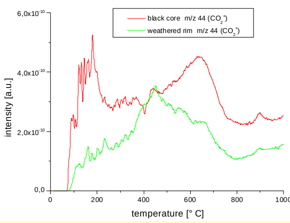


Fig. 9: Formation of CO_2 (m/z 44)

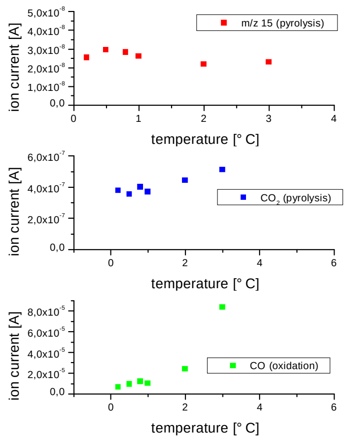


Fig. 9: Pyrolysis parameters as a function of distance to the weathered surface

CONCLUSIONS

Weathering of organic matter in black shales is usually a slow surface dependent process changing composition and content of organic matter. Deeply bleached shales show a strong decrease in organic matter which is not pyrolysable (Fig. 7) and a slight increase in carbon dioxide degassing during pyrolysis (Fig. 9). Unweathered shales from the same sampling area show no carbon dioxide release in that temperature range at all.
In contrast, the amount of hydrocarbons generated by pyrolysis increases with increasing weathering (Fig.6). Obviously those compounds are attached to minerals. This fits very well with the dehydration of illite in that temperature range (Fig. 8). Further studies deal with organic matter isolates (HF-treatment) to estimate the role of minerals in such processes.

Lit.: Heide, K., Gerth, K., Hartmann, E., The detection of an inorganic hydrocarbon formation in silicate melts by means of a direct-coupled-evolved-gas-analysis-system (DEGAS), Thermochimica Acta, 354 (2000) 1-2, 165-172
Schmidt, C.M., Heide, K., Thermal analysis of hydrocarbons in Paleozoic black shales, Journal of thermal analysis and calorimetry, 64 (2001), 1297-1302